

Lab 2 – TrashTag Product Specification

Team Iron

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CS411W

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Contents

3.	Product Requirements.....	3
3.1.	Functional Requirements	3
3.1.1.	Litter Reporting (O: Sara).....	3
3.1.2.	Mapping (O: Sara)	4
3.1.3.	Client Authorization (O: Adam)	4
3.1.4.	Cleanup Tracking (O: Emily)	5
3.1.5.	Verification (O: Emily).....	5
3.1.6.	Community Impact (O: Luca).....	6
3.1.7.	Accessible Resources (O: Adam)	7
3.1.8.	Map Embedding (O: Teddy).....	7
3.1.9.	Analytics (O: Teddy)	7
3.1.10.	Event Coordination (O: Scott)	9

3. Product Requirements

3.1. Functional Requirements

3.1.1. Litter Reporting (O: Sara)

3.1.1.1. Report Litter Location

- 3.1.1.1.1. The system shall automatically request and gather the user's latitude and longitude coordinates through the browsers built in geolocation tool.
- 3.1.1.1.2. The system shall make the users GPS coordinates visible on the live-interactive map and be displayed below the screen for user confirmation and verification before report submission.

3.1.1.2. Upload Photos

- 3.1.1.2.1. The system shall incorporate the separate reusable Camera Capture component into the report screen to allow the user to either capture a picture directly through the camera or upload from gallery.
- 3.1.1.2.2. The system shall also incorporate the reusable Camera Capture component in the Detail screen as it solicits optional photo upload to verify cleanup status.
- 3.1.1.2.3. The system shall accept form report submission without image upload since photo is optional.

3.1.1.3. Mark Trash Size (lbs)

- 3.1.1.3.1. The system shall display the size input field where the user can enter the estimated weight of the litter in pounds with the purpose of providing more detailed information on the littered site.
- 3.1.1.3.2. The system shall accept submission without the size input field, which is optional.

3.1.1.4. Request Cleanup Verification status

- 3.1.1.4.1. The system shall display an "I cleaned this" button on the detail screen to allow users to mark the reported area as cleaned which will correspondingly update the time and date for the report.
- 3.1.1.4.2. The system shall display "Still there" and "cleaned up" buttons on the detail screen to allow users to update the precise status of the reported area by selecting the current status of the reported area.
- 3.1.1.4.3. The system shall display the current status of the litter report on the Trash Detail Screen, reflecting whether the report has

been submitted as “I cleaned this”, “Still there” or “cleaned up” through the verification process.

- 3.1.1.4.4. The system shall allow authenticated users to perform status verification on the detail screen by selecting one of the three buttons and upload optional photo evidence as proof before status update submission.

3.1.2. Mapping (O: Sara)

3.1.2.1. View Litter Map

- 3.1.2.1.1. The system shall integrate the React Leaflet mapping library along with map tiles sourced from OpenStreetMap for the Litter map screen.
- 3.1.2.1.2. The system shall display a leaflet popup marker that shows a photo thumbnail of litter and date report that links to a full detail screen of that report when clicked.
- 3.1.2.1.3. The system shall allow zoom in and zoom out capabilities to view litter reports in proximity.
- 3.1.2.1.4. The system shall display litter reports as trash icons where yellow is for unverified reports, while red is for verified reports where users stated the trash is still there, and green is for reports that have been cleared.

3.1.2.2. Filter by Status

- 3.1.2.2.1. The system shall allow filtering by report status which would either display or hide reports depending on whether they have been verified as cleaned or still there.

3.1.2.3. Filter by Location

- 3.1.2.3.1. The system shall allow users to filter litter reports by specifying a desired proximity range from their current latitude and longitude coordinates.

3.1.3. Client Authorization (O: Adam)

3.1.3.1. Create New Account

- 3.1.3.1.1. The system shall provide the capability for a user to create a new account by submitting a valid email address and password
- 3.1.3.1.2. The system shall verify that the provided email address is not already associated with an existing account

3.1.3.2. Login

- 3.1.3.2.1. The system shall provide the capability for a registered user to login using a valid username and password
- 3.1.3.2.2. The system shall deny access and notify the user if the provided credentials are invalid.

- 3.1.3.2.3. The system shall keep the user logged in for their current session once they successfully login.
- 3.1.3.2.4. The system shall verify the provided credentials from the user against all users' data in the database before granting access.

3.1.3.3. View and Edit Profile

- 3.1.3.3.1. The system shall provide the capability for a user to view their stored profile information
- 3.1.3.3.2. The system shall provide the capability for a user to update their profile information, including display name, location, and profile image.
- 3.1.3.3.3. The system shall update profile information in the SQLite database upon the user's successful submission of a profile update.

3.1.4. Cleanup Tracking (O: Emily)

3.1.4.1. Update Cleanup Status

- 3.1.4.1.1. The system shall provide the user with a status update regarding trash cleanup.
- 3.1.4.1.2. The system shall show the full details of a trash report with a corresponding image gallery.
- 3.1.4.1.3. The system shall fetch the trash report details from the created API.

3.1.4.2. Upload Cleanup Proof

- 3.1.4.2.1. The system shall allow the user to upload multiple images of trash to the trash report.

3.1.4.3. Monitor Cleanup Status

- 3.1.4.3.1. The system shall allow the user to monitor cleanup status by viewing the trash report's updates in real time.
- 3.1.4.3.2. The system shall allow the user to refresh the trash report page and view new updates upon refresh.

3.1.5. Verification (O: Emily)

3.1.5.1. Verify Accurate Cleanup Status

- 3.1.5.1.1. The system shall provide the user with the ability to verify trash reports once the affected area has been cleaned.
- 3.1.5.1.2. The system shall provide the user with the ability to verify reports by clicking if the trash is still active or if it has been cleaned.
- 3.1.5.1.3. The system shall provide the user with the ability to add optional photos if the trash has been either cleaned up or not.

3.1.5.2. Verify Cleanup Completion

- 3.1.5.2.1. The system shall provide the user with a notification if the verified report has been marked as “completed”.
- 3.1.5.2.2. The system shall allow the user to confirm whether the report has been verified.

3.1.5.3. Flag Suspicious Content

- 3.1.5.3.1. The system shall provide the capability for the user to flag suspicious content.
- 3.1.5.3.2. The system shall allow the user to flag spam trash reports.
- 3.1.5.3.3. The system shall allow the user to submit suspicious content to moderators for review.
- 3.1.5.3.4. The system shall allow the user to submit spam trash reports to moderators for review.

3.1.6. Community Impact (O: Luca)

3.1.6.1. View Impact Statistics

- 3.1.6.1.1. The system shall provide the user with basic statistics related to the litter reported such as weight in pounds

3.1.6.2. Gamifications/Rewards

- 3.1.6.2.1. The system shall reward users with points based on their interactions with the application.
- 3.1.6.2.2. The system shall award users points based on this system.
- 3.1.6.2.3. For submitting a verified photo of litter with its location, users will earn 100 points.
- 3.1.6.2.4. For verifying litter is still there, users will earn 50 points.
- 3.1.6.2.5. For verifying trash has been cleaned up, users will earn 50 points.
- 3.1.6.2.6. For creating a verified cleanup event that comes to fruition, users will earn 200 points.
- 3.1.6.2.7. For joining a cleanup event and arriving onsite, users will earn 100 points.
- 3.1.6.2.8. For completing a verified cleanup event that the user created, users will earn 500 points.
- 3.1.6.2.9. For completing a verified cleanup event that the user joined, users will earn 200 points.

3.1.6.3. Top Contributors

- 3.1.6.3.1. The system shall provide users with the option to compete on four leaderboards.
 - 3.1.6.3.1.1. Local weekly.
 - 3.1.6.3.1.2. Local overall.
 - 3.1.6.3.1.3. Global weekly.
 - 3.1.6.3.1.4. Global overall.

- 3.1.6.3.2. The system shall omit the local leaderboards if users do provide their hometown location.
- 3.1.6.3.3. The system shall calculate ranking based on points.
- 3.1.6.3.4. The system shall refresh weekly leaderboards on Sunday at 0:00 EST.

3.1.7. Accessible Resources (O: Adam)

3.1.7.1. Local laws and resources

- 3.1.7.1.1. The system shall provide the capability for users to view environmental resources specific to their detected location including local laws, resources, and local cleanup agencies.
- 3.1.7.1.2. The system shall retrieve resource information based on the user's detected geographic location.
- 3.1.7.1.3. The system shall provide the user with reliable resources from trustworthy sources.
- 3.1.7.1.4. The system shall display the resource information within the application interface in a readable format.
- 3.1.7.1.5. The system shall allow users to access resource information without requiring logging in.

3.1.8. Map Embedding (O: Teddy)

3.1.8.1. Embed Map Widget (Partial)

- 3.1.8.1.1. The system shall provide an embeddable map view that displays stored trash report locations on an external organization or government website.
- 3.1.8.1.2. The system shall display each trash report in the embedded map view using the geographic coordinates stored for that report.
- 3.1.8.1.3. The system shall display the selected report's location and cleanup status when a user selects a report marker in the embedded map view.
- 3.1.8.1.4. The system shall restrict the embedded map view to report data designated for public display.

3.1.9. Analytics (O: Teddy)

3.1.9.1. Highlight Predicted Litter Trends (Deferred)

- 3.1.9.1.1. The system shall calculate litter trend results using stored trash report timestamps and geographic locations.
- 3.1.9.1.2. The system shall group trend results by a defined time period and geographic area.
- 3.1.9.1.3. The system shall identify geographic areas in which report counts increase over successive time periods.

- 3.1.9.1.4. The system shall display predicted trend results separately from current reported litter locations.

3.1.9.2. Download Reports (Partial)

- 3.1.9.2.1. The system shall allow an authorized organization or government user to export stored trash report data.
- 3.1.9.2.2. The system shall include, at minimum, report identifier, location, submission timestamp, and cleanup status in each exported report file.
- 3.1.9.2.3. The system shall generate the exported report file from the current data stored in the system database at the time of the request.
- 3.1.9.2.4. The system shall deny report export requests from unauthenticated users.

3.1.9.3. View Density - Heavy Litter Analytics (Deferred)

- 3.1.9.3.1. The system shall calculate litter density as the number of stored trash reports within a defined geographic area.
- 3.1.9.3.2. The system shall generate a density view that distinguishes areas of higher litter concentration from areas of lower litter concentration.
- 3.1.9.3.3. The system shall update density results when new trash reports are added or existing reports are removed from active status.
- 3.1.9.3.4. The system shall allow an authorized user to view density results for a selected map region.

3.1.9.4. Track Event Impact (Deferred)

- 3.1.9.4.1. The system shall allow authorized users to associate cleanup events with one or more trash reports or geographic cleanup areas.
- 3.1.9.4.2. The system shall track changes in report status associated with a cleanup event.
- 3.1.9.4.3. The system shall provide summary metrics for a cleanup event, including the number of reports affected and the number of reports marked cleaned.
- 3.1.9.4.4. The system shall allow authorized users to view event impact information after cleanup event completion.

3.1.9.5. View Cleanup Statistics (Partial)

- 3.1.9.5.1. The system shall provide users with cleanup statistics derived from stored report and cleanup status data.
- 3.1.9.5.2. The system shall display, at minimum, the number of reports submitted, the number of reports marked cleaned, and the number of reports remaining active.

3.1.9.5.3. The system shall generate cleanup statistics from data stored in the system database.

3.1.9.5.4. In the prototype, cleanup statistics may be limited to basic aggregate values rather than advanced analytics.

3.1.10. Event Coordination (O: Scott)

3.1.10.1. Coordinate Cleanup Events

3.1.10.1.1. The system shall allow users to create events with a given title, description, location, date, and time. Optional photos may also be included.

3.1.10.2. Claim Cleanup Area

3.1.10.2.1. Organization type user accounts can mark arbitrary areas (up to 5 square miles) as areas of interest and commitment to keeping clean. They then receive alerts when new trash reports are created in that area.

3.1.10.3. Delegate Cleanup Tasks

3.1.10.3.1. Organization type user accounts can assign other normal user accounts trash reports.

3.1.10.4. Event Participation

3.1.10.4.1. Users can view on their account page which events they have marked interest or commitment in.

3.1.10.4.2. Events show which users have marked interest or commitment in.

3.1.10.5. View Cleanup Events

3.1.10.5.1. An events list page is available that shows all events within a threshold distance. Threshold is settable by the user in 1, 3, 5, and 10 miles.

3.1.10.6. RSVP to Events

3.1.10.6.1. Users may mark interest or commitment to events.

3.1.10.7. Check-In to Events

3.1.10.7.1. Users can mark that they are currently attending an event after the event time has started and they are within 500ft of the event location.